Offline Reinforcement Learning with Implicit State-State Planning (Supplementary Material)

Shuo Cao, Yuhu Cheng, Xuesong Wang

Appendix A. Proof for Theorem 1

Proof: Consider two distinct Q-functions, $\hat{Q}(s,a)$ and $\hat{Q}'(s,a)$. According to the definition of $\hat{\mathcal{B}}^{\pi_g}$, its infinite norm operation satisfies

$$
\begin{aligned}
&\left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q})(s,a)-(\hat{\mathcal{B}}^{\pi_g}\hat{Q}')(s,a)\right\|_\infty\\
=&\max_{s,a}\left|(r+\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}(s',a'_{IS})])-(r+\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}'(s',a'_{IS})])\right|\\
\leq&\max_{s'}\left|\gamma\mathbb{E}_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}[\hat{Q}(s',a'_{IS})-\hat{Q}'(s',a'_{IS})]\right|\\
\leq&\gamma\max_{s',a'_{IS}\sim\mathcal{I}(s',\mathcal{X}_g(s'))}\left|\hat{Q}(s',a'_{IS})-\hat{Q}'(s',a'_{IS})\right|\\
\leq&\gamma\max_{s,a}\left|\hat{Q}(s,a)-\hat{Q}'(s,a)\right|\\
=&\gamma\left\|\hat{Q}(s,a)-\hat{Q}'(s,a)\right\|_\infty.
\end{aligned}
\tag{A.1}
$$

According to Eq. (A.1), $\hat{\mathcal{B}}^{\pi_g}$ is a γ-contraction mapping, which satisfies $\left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q})-(\hat{\mathcal{B}}^{\pi_g}\hat{Q}')\right\|_\infty \leq \gamma\left\|\hat{Q}-\hat{Q}'\right\|_\infty$. Setting $\hat{Q}^*$ as the unique fixed point of the Q-function $\hat{Q}$, then repeatedly performing $\hat{\mathcal{B}}^{\pi_g}$ yields

$$
\begin{aligned}
\left\|\hat{Q}_{k+1}-\hat{Q}^*\right\|_\infty &= \left\|(\hat{\mathcal{B}}^{\pi_g}\hat{Q}_k)-(\hat{\mathcal{B}}^{\pi_g}\hat{Q}^*)\right\|_\infty\\
&\leq \gamma\left\|\hat{Q}_k-\hat{Q}^*\right\|_\infty \leq \cdots \leq \gamma^{k+1}\left\|\hat{Q}_0-\hat{Q}^*\right\|_\infty.
\end{aligned}
\tag{A.2}
$$

Eq. (A.2) indicates that for $\gamma < 1$, as the number of iterations k increases, $\hat{Q}_{k+1}$ converges to $\hat{Q}^*$, i.e., $\lim_{k\to\infty}\hat{Q}_{k+1}=\hat{Q}^*$. Thus, Theorem 1 is proofed. ■

Appendix B. Proof for Theorem 2

Proof: Under Assumption 1, solving the deterministic MDP through explicit state-stitching can be regarded as solving a sub-MDP via explicit action-stitching. In this sub-MDP, only the mappings $\mathcal{I}(s, s')$ are included, where $\mathcal{I}(s, s')$ returns a set of feasible actions for each state s, this is precisely the mechanism we refer to as implicit state-stitching. Specifically, for any given state s, we have

$$\hat{Q}(s, a) \geq \max_{s'} \hat{Q}(s, s') = \max_{s'} \hat{Q}(s, a = \mathcal{I}(s, s')) \quad \text{for least one } a. \tag{B.1}$$

Intuitively, this means that there exists at least one action a for which the Q-value in the original MDP is no smaller than the maximal $\hat{Q}$-value obtained by stitching over possible successor states s'.

Although the actions generated by $\mathcal{I}(s, s')$ may only constitute a subset of the full action space of the original MDP, we assume that the chosen successor state s' is optimal. Under this assumption, the return associated with any action in the original MDP cannot exceed the return achieved through the stitched action sequence provided by $\mathcal{I}(s, s')$. Formally, for each s we have

$$\hat{Q}(s, a) \leq \max_{s'} \hat{Q}(s, \mathcal{I}(s, s')) \quad \text{for all } a. \tag{B.2}$$

Combining Eqs. (B.1) and (B.2), we conclude that for a deterministic MDP, there exist

$$\max_{a} \hat{Q}(s, a) = \max_{s'} \hat{Q}(s, s') = \max_{s'} \hat{Q}(s, a = \mathcal{I}(s, s')). \tag{B.3}$$

This result demonstrates that, in a deterministic MDP, the maximization over actions a in the original Q-function is equivalent to the maximization over successor states s' in the implicit state-stitching formulation. Hence, Theorem 2 is proven. ■

Appendix C. Proof for Theorem 3

Proof: Recalling Eq. (19) and applying the triangle inequality, we obtain

$$\begin{aligned}
&\left\|\hat{Q}^{\pi_g}\left(s,\pi_g(s)\right)\right) - Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\| \\
=&\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \\
=&\Big\|(\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - \hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)) + (\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right) - \hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)) \\
&\quad + (\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right) - \bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)) + (\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right) - Q^{\pi_\beta}\left(s,\pi_\beta(s)\right))\Big\| \\
\leq&\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - \hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\| + \left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right) - \hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \\
&+ \left\|\hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right) - \bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| + \left\|\bar{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right) - Q^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\|. \qquad \text{(C.1)}
\end{aligned}$$

Considering $\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - \hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\|$, and given the K_Q-Lipschitz as stated in Assumption 2, we can derive

$$\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - \hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\| \leq K_Q\left\|\mathcal{I}(s,\mathcal{X}_g(s)) - \pi_\beta(s)\right\|. \qquad \text{(C.2)}$$

Based on $\pi_\beta(s) = \mathcal{I}(s,s')$ and the K_I-Lipschitz as stated in Assumption 2, $\left\|\mathcal{I}(s,\mathcal{X}_g(s)) - \pi_\beta(s)\right\| = \left\|\mathcal{I}(s,\mathcal{X}_g(s)) - \mathcal{I}(s,s')\right\| \leq K_{\mathcal{I}}\left\|\mathcal{X}_g(s) - s'\right\|$ holds. Recalling the state constraint threshold $\left\|\mathcal{X}_g(s) - s'\right\| \leq \epsilon_g$ from Eq. (12), there exist

$$\left\|\mathcal{I}(s,\mathcal{X}_g(s)) - \pi_\beta(s)\right\| \leq K_{\mathcal{I}}\epsilon_g. \qquad \text{(C.3)}$$

Combining Eqs. (C.2) and (C.3), we obtain

$$\left\|\hat{Q}^{\pi_g}\left(s,\mathcal{I}(s,\mathcal{X}_g(s))\right) - \hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right)\right\| \leq K_Q K_{\mathcal{I}}\epsilon_g. \qquad \text{(C.4)}$$

Next, considering $\left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right) - \hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\|$, and based on the value gaps under different policies as presented in Lemma 1, it follows that

$$\left\|\hat{Q}^{\pi_g}\left(s,\pi_\beta(s)\right) - \hat{Q}^{\pi_\beta}\left(s,\pi_\beta(s)\right)\right\| \leq \left\|\hat{Q}^{\pi_g}(s,a) - \hat{Q}^{\pi_\beta}(s,a)\right\| \leq R_{\max}\sqrt{2K_{\mathcal{I}}\epsilon_g}/(1-\gamma)^2. \qquad \text{(C.5)}$$

Due to the estimation deviation between the $\hat{Q}^{\pi_\beta}$ estimated by off-policy expectile regression and the $\bar{Q}^{\pi_\beta}$ estimated by MSE, we define the upper bound of the bias between them as δ_τ:

$$\left\|\hat{Q}^{\pi_\beta}\left(s, \pi_\beta(s)\right) - \bar{Q}^{\pi_\beta}\left(s, \pi_\beta(s)\right)\right\| \le \left\|\hat{Q}^{\pi_\beta}(s, a) - \bar{Q}^{\pi_\beta}(s, a)\right\| \le \delta_\tau. \tag{C.6}$$

Furthermore, based on the sampling error result from Lemma 2, we can conclude that

$$\left\|\bar{Q}^{\pi_\beta}\left(s, \pi_\beta(s)\right) - Q^{\pi_\beta}\left(s, \pi_\beta(s)\right)\right\| \le \left\|\bar{Q}^{\pi_\beta}(s, a) - Q^{\pi_\beta}(s, a)\right\| \le \delta_{\mathcal{D}}. \tag{C.7}$$

Finally, substituting Eqs. (C.4)-(C.7) into Eq. (C.1), we can derive

$$\left\|\hat{Q}^{\pi_g}\left(s, \mathcal{I}(s, \mathcal{X}_g(s))\right) - Q^{\pi_\beta}\left(s, \pi_\beta(s)\right)\right\| \le \delta_{\mathcal{D}} + \delta_\tau + K_Q K_{\mathcal{I}} \epsilon_g + R_{\max}\sqrt{2K_{\mathcal{I}}\epsilon_g}/(1-\gamma)^2. \tag{C.8}$$

Thus, Theorem 3 is proofed. ■

Appendix D. Proof for Theorem 4

Proof: In ORL, once the training dataset is fixed, the performance gap $\varepsilon_{\mathcal{D}}$ between the actual global optimal policy π^* and the local behavior policy π_β derived from the finite dataset, becomes fixed:

$$\varepsilon_{\mathcal{D}} = J(\pi^*) - J(\pi_\beta). \tag{D.1}$$

While the performance gap between execution-policy π_e and optimal policy π^* is $J(\pi^*) - J(\pi_e)$, by applying Eq. (22) to extend $|J(\pi^*) - J(\pi_e)|$ and using the triangle inequality, we obtain

$$\begin{aligned} |J(\pi^*) - J(\pi_e)| &= \left|\left(J(\pi^*) - J(\pi_g)\right) + \left(J(\pi_g) - J(\pi_\beta)\right) + \left(J(\pi_\beta) - J(\pi_e)\right)\right| \\ &\le \left|J(\pi^*) - J(\pi_g)\right| + \left|J(\pi_g) - J(\pi_\beta)\right| + \left|J(\pi_\beta) - J(\pi_e)\right|. \end{aligned} \tag{D.2}$$

Considering $\left|J(\pi^*) - J(\pi_g)\right|$, based on the definition of $J(\pi)$ in Section 2, Lemma 3, and Eq. (D.1), we can derive

$$\begin{aligned}
\left|J(\pi^*) - J(\pi_g)\right| =& \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi^*}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_g}(s)}\left[r(s)\right]\right| \\
\leq& \frac{1}{1-\gamma}\left|\int_s \left(d^{\pi^*}(s) - d^{\pi_g}(s)\right) r(s)\mathrm{d}s\right| \\
\leq& \frac{R_{\max}}{1-\gamma}\int_s \left|d^{\pi^*}(s) - d^{\pi_g}(s)\right|\mathrm{d}s \\
\leq& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \max_s \left\|\mathcal{X}^*(s) - \mathcal{X}_g(s)\right\| \\
=& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \max_s \left\|\mathcal{X}^*(s) - s' + s' - \mathcal{X}_g(s)\right\| \\
\leq& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \max_s \left(\left\|\mathcal{X}^*(s) - s'\right\| + \left\|s' - \mathcal{X}_g(s)\right\|\right) \\
\leq& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} (\varepsilon_{\mathcal{D}} + \epsilon_g).
\end{aligned} \tag{D.3}$$

By considering $\left|J(\pi_g) - J(\pi_\beta)\right|$ and applying Lemma 3, we obtain

$$\begin{aligned}
\left|J(\pi_g) - J(\pi_\beta)\right| =& \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_g}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\beta}(s)}\left[r(s)\right]\right| \\
\leq& \frac{1}{1-\gamma}\left|\int_s (d^{\pi_g}(s) - d^{\pi_\beta}(s)) r(s)\mathrm{d}s\right| \\
\leq& \frac{R_{\max}}{1-\gamma}\int_s |d^{\pi_g}(s) - d^{\pi_\beta}(s)|\mathrm{d}s \\
\leq& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \max_s \left\|\mathcal{X}_g(s) - s'\right\| \\
\leq& \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \epsilon_g.
\end{aligned} \tag{D.4}$$

Similarly, by considering $\left|J(\pi_\beta) - J(\pi_e)\right|$ and utilizing Lemma 3, we obtain

$$
\begin{aligned}
\left|J(\pi_\beta) - J(\pi_e)\right| &= \left|\frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\beta}(s)}\left[r(s)\right] - \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_e}(s)}\left[r(s)\right]\right| \\
&\leq \frac{1}{1-\gamma}\left|\int_s \left(d^{\pi_\beta}(s) - d^{\pi_e}(s)\right) r(s)\mathrm{d}s\right| \\
&\leq \frac{R_{\max}}{1-\gamma}\int_s \left|d^{\pi_\beta}(s) - d^{\pi_e}(s)\right| \mathrm{d}s \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \max_s \left\|s' - \mathcal{X}_e(s)\right\| \\
&\leq \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} \epsilon_e.
\end{aligned} \tag{D.5}
$$

Substituting Eqs. (D.3)-(D.5) into Eq. (D.2), the performance lower bound of execution-policy π_e is

$$
|J(\pi^*) - J(\pi_e)| \leq \frac{R_{\max}}{1-\gamma} C K_T K_{\mathcal{I}} (\varepsilon_{\mathcal{D}} + 2\epsilon_g + \epsilon_e). \tag{D.6}
$$

Thus, Theorem 4 is proofed.

Appendix E. Dataset Description and Hyper-parameter Setting

Table E.1: Description and min–max return references for MuJoCo, AntMaze, and CityLearn environments.

Control tasks	Learning goal	Reward	Ref. min	Ref. max
HalfCheetah	Control a cheetah-like robot to run as quickly forward as possible	Dense	-280.18	12135.0
Hopper	Control a monopod-like robot to jump as far forward as possible	Dense	-20.27	3234.3
Walker2d	Control a biped-like robot to walk as quickly forward as possible	Dense	1.63	4592.3
Antmaze-umaze	Control an 8-DOF ant-like robot to find the shortest path in a simple-maze layout	Sparse	0.0	1.0
Antmaze-medium	Control an 8-DOF ant-like robot to find the shortest path in a medium-maze layout	Sparse	0.0	1.0
Antmaze-large	Control an 8-DOF ant-like robot to find the shortest path in a difficult-maze layout	Sparse	0.0	1.0
CityLearn	Coordinate domestic hot water and chilled water storage in electricity-consuming buildings to reshape the overall electricity demand profile	Dense	16280	50350

Appendix F. Quantitative Evaluation and Stability Analysis of the Inverse Dynamics Model

To address **Q5** and the concerns regarding IDM reliability and its coupling effect on policy learning, we present a quantitative study on IDM convergence behavior and its influence on Q-value estimation and policy performance. Furthermore, we provide

Table E.2: Dataset types across MuJoCo, AntMaze, and CityLearn environments.

Dataset type	**Description of collected data quality/policy**	**Policy num.**	**Policy mix**	**Experiences**
Random	Random-level policy	Single	100%	10^6
Medium	Medium-level policy that achieves one-third of expert policy performance	Single	100%	10^6
Medium-replay	Medium dataset and replay buffer during medium-level agent training	Multiple	$50\% : 50\%$	$\approx 2 \times 10^6$
Medium-expert	Medium-level policy and expert-level policy	Multiple	$50\% : 50\%$	2×10^6
Expert	Expert-level agent policy or demonstration	Single	100%	10^6
Default	Goal-reaching policy from a fixed location to a specific goal	Single	100%	10^6
Diverse	Goal-reaching policy from a random location to a random goal	Single	100%	10^6
Play	Starting from hand-picked location set to specific hand-picked goal set	Single	100%	10^6
Dataset type	**Description of collected data quality/policy**	**Policy num.**	**Policy mix**	**Trajectories**
Low	Collecting policy achieving 25% of expert return, highly suboptimal	Single	100%	$\{99, 999, 9999\}$
Medium	Collecting policy achieving 50% of expert return, moderately suboptimal	Single	100%	$\{99, 999, 9999\}$
High	Collecting policy achieving 75% of expert return, near-expert	Single	100%	$\{99, 999, 9999\}$

Table E.3: Network structure and hyperparameter setting of ISSP.

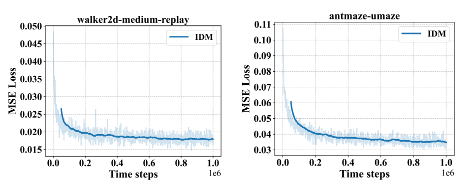
	Hyper-parameter	**Value**
Network	Optimizer	Adam
	Mini-batch size	256
	Activation function	ReLU
	Critic/STM/IDM network hidden layer	4
	Critic/STM/IDM each hidden layer unit	256
	Critic/STM/IDM network learning rate	3×10^{-4}
	All target network update rate v	5×10^{-3}
ISSP	Discount factor γ	0.99
	State Exploring noise σ	0.05
	Maximum training time step T	1×10^6
	STM delay update frequency d	2
	Off-policy expectile regression weight $\{\tau_1, \tau_2\}$	{0.2, 0.3} for MuJoCo and CityLearn {0.9, 0.9} for AntMaze
	Balance coefficient α	0.4
	Mild constraint factor λ	10 for MuJoCo and CityLearn 2.5 for AntMaze

a mechanism-level discussion on how the architectural design of ISSP contributes to stability and convergence efficiency.

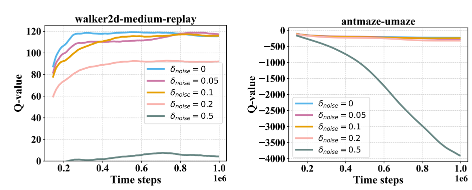
1) IDM Training Quality and Convergence: As shown in Fig. F.1(a), the MSE loss of IDM on both walker2d-medium-replay and antmaze-umaze environments converges rapidly within the first 0.2×10^6 steps and remains stable throughout the training process. The final MSE stabilizes at a low range (0.015-0.035), indicating that the IDM learns an accurate and well-conditioned mapping from state transitions to actions.

2) Impact of IDM Prediction Error on Q-value Estimation and Policy Performance: To quantitatively evaluate the sensitivity of ISSP to IDM inaccuracies, we inject Gaussian noise into the IDM output during both training and inference: $a_{\text{noise}} = \mathcal{I}(s, s') + \delta_{\text{noise}} \cdot \mathcal{N}(0, I)$, where $\delta_{\text{noise}} \in \{5\%, 10\%, 20\%, 50\%\} = \{0.05, 0.1, 0.2, 0.5\}$. From Figs F.1(b)-(c), we can draw the following two conclusions. (i) Q-value Estimation: In walker2d-medium-replay, Q-values remain stable under noise levels up to 0.2 and degrade only under extreme noise 0.5. In antmaze-umaze, large noise leads to underestimation, whereas moderate noise results in only minor deviations, demonstrating the robustness of Q-value estimation. (ii) Policy Performance: Normalized scores exhibit strong resilience to IDM perturbations. Notably, in antmaze-umaze, even with 10% noise ($\delta_{\text{noise}} = 0.1$), performance remains comparable to the noise-free baseline. This indicates that STM planning compensates for mild IDM inaccuracies by state perturbation smoothing.

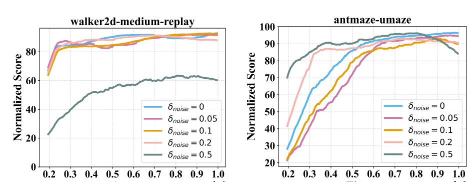
3) Discussion on Stability and Convergence Efficiency: Although ISSP involves interactions between the STM and IDM, the learning architecture remains weakly coupled and regularized, which promotes stable optimization and improved convergence efficiency. (i) Structured Search Space: By planning in the state-space rather than the raw action-space, the STM exploits the continuity and smoothness of underlying physical dynamics, thereby reducing optimization variance compared with direct action-space search. (ii) Decoupled Action Grounding: The IDM provides a data-consistent mapping from transitions to actions and is treated as a fixed functional transformation during STM optimization. This prevents gradient-level feedback between the two models, avoiding mutual error amplification and enhancing training stability. (iii) Implicit Feasibility Regularization: Since the IDM maps transitions to realizable in-dataset actions, STM optimization is implicitly restricted to feasible transitions. This prunes transition-infeasible, effectively reducing the search space and accelerating convergence. (iv) Robustness via State Smoothing Regularization: The smoothed perturbations introduced for IDM training in Eq. (14) enforce locally robust state-action mappings under small state variations. This stochastic smoothing propagates into Q-function evaluation in Eq. (7), effectively smoothing the induced value distribution and mitigating abrupt value fluctuations in OOD regions, thereby reducing Q-value overestimation and oscillatory updates. Together, these mechanisms transform policy learning into a structured and regularized state-transition search process, which empirically leads to faster and more stable convergence in Figs. H.3-H.4.



(a) IDM convergence curves



(b) Q-value estimation under IDM prediction errors

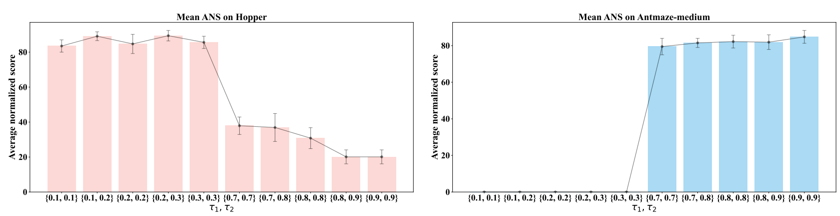


(c) Policy performance under IDM prediction errors

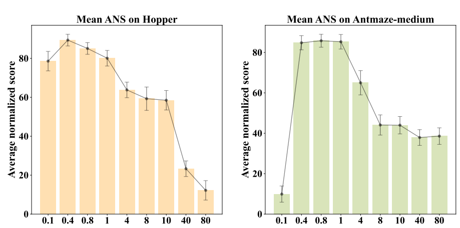
Fig. F.1. IDM convergence, Q-value estimation, and policy performance under different levels of IDM prediction error.

Appendix G. Hyperparameter Analysis and Selection Insights

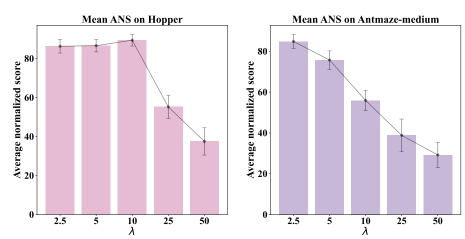
To further strengthen the rigor of this study and address **Q6**, we conduct a sensitivity analysis on key hyperparameters: $\{\tau_1, \tau_2\}$, α, and λ. Following standard practice in ORL, these hyperparameters are not tuned separately for each task. Instead, we adopt a category-wise offline tuning protocol, where a single representative environment is used to select hyperparameters that are then fixed for all tasks within the same domain. Specifically, Hopper and AntMaze-medium are chosen as representative environments for dense-reward (MuJoCo/CityLearn) and sparse-reward (AntMaze) domains, respectively. All tuning is conducted strictly offline using the average normalized score on training dataset as the selection criterion. Once chosen, the hyperparameters are applied uniformly across tasks in the corresponding category, as summarized in Fig. G.2 and Table E.3.



(a) Off-policy expectile weight $\{\tau_1, \tau_2\}$



(b) Balance coefficient α



(c) Mild constraint factor λ

Fig. G.2. Decision-making quality of ISSP under varying off-policy expectile regression weight $\{\tau_1, \tau_2\}$, balance coefficient α, and mild constraint factor λ, on the Hopper and Antmaze-medium environments.

1) Off-policy Expectile Weight $\{\tau_1, \tau_2\}$: These parametersjointly control the quantile regression targets of $\hat{Q}$ toward $(\hat{\mathcal{B}}^{\pi_g}\hat{Q})$ during training, thereby strengthening the in-dataset state-action correspondence of STM and amplifying the influence of actions predicted by STM with the aid of IDM on future rewards. Fig. G.2(a) shows strong category-level consistency. For the dense-reward Hopper environment, ANS remains stable and high for low expectile weights $\{\tau_1, \tau_2\} < 0.5$. We recommend $\{\tau_1, \tau_2\} \in \{0.2, 0.3\}$, which is applied to all 24 dense-reward tasks. This choice maintains value stability by filtering potential overestimation in high-frequency feedback environments. In contrast, for the sparse-reward AntMaze-medium environment, higher weights $\{\tau_1, \tau_2\} > 0.5$ are necessary to capture rare success signals. Increasing these parameters significantly improves ANS, whereas low values cause failures. We therefore fix $\{\tau_1, \tau_2\} = \{0.9, 0.9\}$ for all 6 sparse-reward tasks. This categorical selection demonstrates that ISSP does not require per-task tuning once the reward density is identified.

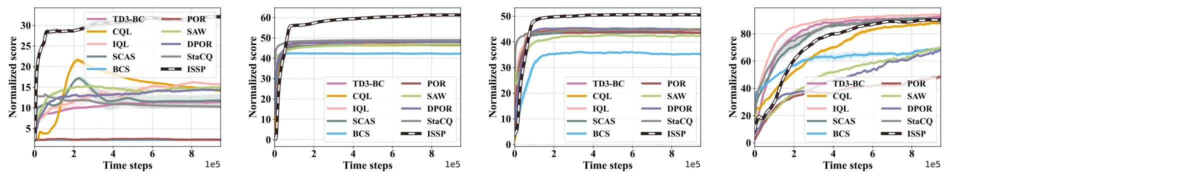
2) Balance Coefficient α: For ISG-PI and MISE-PI, α regulates the trade-off between the mode-seeking objective (maximizing $\hat{Q}$) and the mode-covering property of BCS. Fig. G.2(b) shows a consistent trend: ANS initially increases with α but decreases after exceeding a threshold, indicating that a balanced trade-off is essential. We set $\alpha = 0.4$ as a robust default across all domains.

3) Mild Constraint Factor λ: The factor λ, trained independently within MISE-PI and applied only during inference, regularizes the execution-policy through a mild state mode-covering constraint. Increasing λ relaxes this constraint and expands the action exploration space induced by the STM and IDM. Sensitivity analysis in Fig. G.2(c) shows that the optimal value is 10 for the Hopper task as well as the dense-reward MuJoCo/CityLearn domain, and 2.5 for the AntMaze-medium task as well as the sparse-reward AntMaze domain.

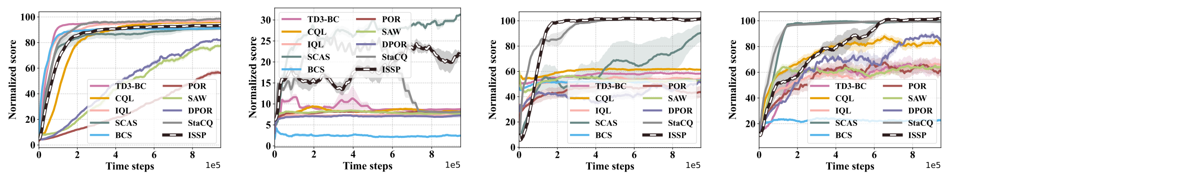
Appendix H. Normalized Score Learning Curve

This section reports the normalized score learning curves of ISSP and baseline methods over one million training steps on D4RL and 0.2 million steps on NeoRL.

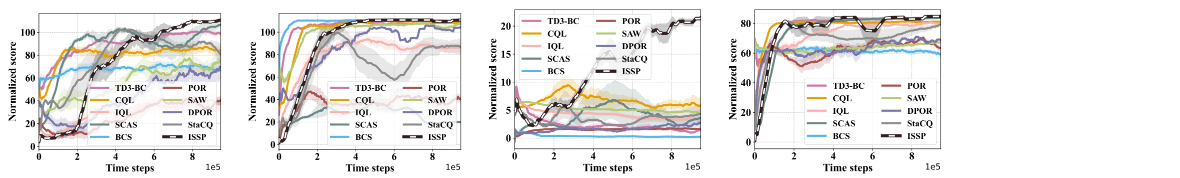
The results illustrate the training dynamics and facilitate a comprehensive evaluation of convergence behavior, performance improvement, and learning stability.



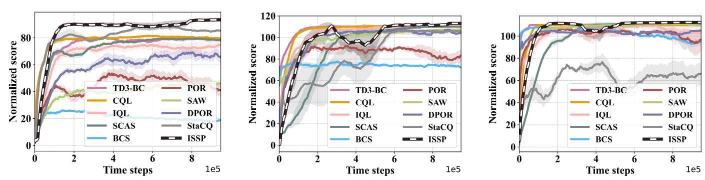
(a) halfcheetah-random (b) halfcheetah-medium (c) halfcheetah-medium-replay (d) halfcheetah-medium-expert



(e) halfcheetah-expert (f) hopper-random (g) hopper-medium (h) hopper-medium-replay

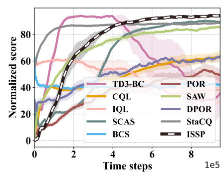


(i) hopper-medium-expert (j) hopper-expert (k) walker2d-random (l) walker2d-medium

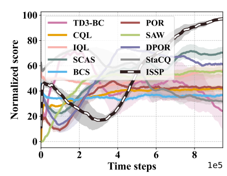


(m) walker2d-medium-replay (n) walker2d-medium-expert (o) walker2d-expert

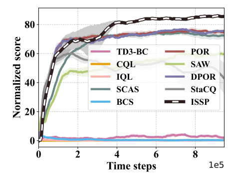
Fig. H.3. Normalized score learning curves of ISSP against ORL baselines on the D4RL MuJoCo dataset tasks. The centerline and its shadow represent the mean and standard deviation across five seeds, respectively. Smoothing via a moving average window is applied to enhance visual clarity.



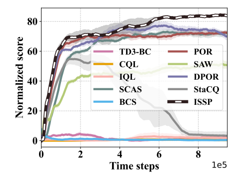
(a) antmaze-umaze



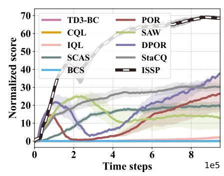
(b) antmaze-umaze-diverse



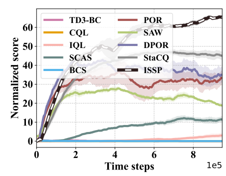
(c) antmaze-medium-play



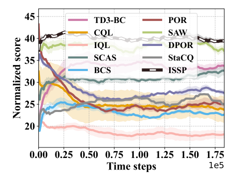
(d) antmaze-medium-diver



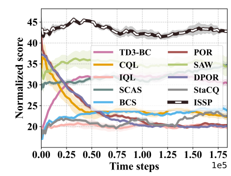
(e) antmaze-large-play



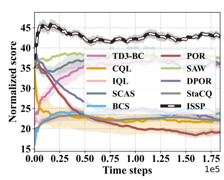
(f) antmaze-large-diverse



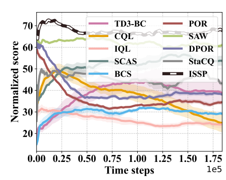
(g) citylearn-low-99



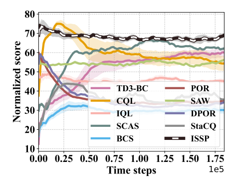
(h) citylearn-low-999



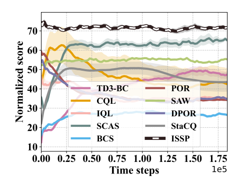
(i) citylearn-low-9999



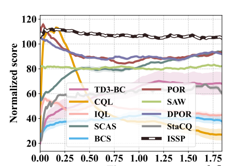
(j) citylearn-medium-99



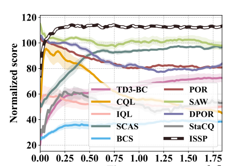
(k) citylearn-medium-999



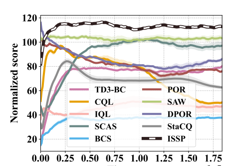
(l) citylearn-medium-9999



(m) citylearn-high-99



(n) citylearn-high-999



(o) citylearn-high-999

Fig. H.4. Normalized score learning curves of ISSP against ORL baselines on the D4RL AntMaze and NeoRL CityLearn dataset tasks.